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Testing Improved Crossing-Based Selection Techniques

Traditional point-and-click methods seem to dominate the user-based pointing interaction paradigms, with almost all applications requiring the use of a mouse and pointer. However, a sometimes neat and overlooked pointing interaction, like the trail-based/crossing-based selection brush, could potentially offer a faster and more dynamic way for users to interact with digital content, providing both precision and efficiency in tasks like selection and drawing.

However, while crossing-based selection offer a hands-free, fluid alternative to pointing, they often suffer from accidental selections, ambiguous visual cues, and unintuitive deselection methods, especially in dense or fast-paced interfaces. Many existing implementations lack a selection delay, causing users to accidentally activate elements simply by passing over them too quickly. This is the case in tools like AutoCAD's crossing selection mode, where selection is triggered instantly when dragging over objects, often requiring unnecessary corrections.

Additionally, deselection mechanisms in some crossing-based systems are counterintuitive, requiring users to re-cross in the opposite direction to undo selections, which can lead to confusion and inefficiency.

My brush is designed to resolve these issues by introducing a more intentional and responsive selection model: one that incorporates a short delay to reduce unintended activations, visual trails for clarity, and easy deselection through right-click interaction. This ensures users

the fluidity of crossing-based input while gaining the accuracy, feedback, and confidence often associated with more deliberate, click-based systems.

To begin with, you can enable/disable my brush interactions (I refer to this state as brush mode), by simply hitting the CTRL key. This will replace the mouse pointer icon with a white circle engraved with a black X. Now in brush mode, to select items the user can simply drag the mouse while pressing the left mouse button, which draws a glowing trail as the user moves across the screen. This is responsible for selecting targets, but only after the user has hovered over an item for a short, configurable delay—mitigating the risk of accidental selections. Additionally, nearby elements within a defined threshold are also automatically grouped and selected, making the process more efficient in clustered layouts. If a user changes their mind or selects something unintentionally, they can simply right-click to deselect any highlighted item as long as they are in ‘brush mode’. These features combine to offer a selection experience that is both flexible and forgiving—allowing users to work quickly without sacrificing precision or control.

Before trying my trail-based brush, I had my participants demo their favorite selection tools from other software they regularly use. This included the lasso tool in Photoshop, and the rectangular marquee tool in Figma and Microsoft Paint. These tools were preferred, according to them, for their simplicity, direct control, and usefulness. The lasso and Procreate brush had a more fluid, natural feel, making it easy to trace irregular shapes according to tom my participants. The marquee was chosen due to its precision and clear feedback, with selections immediately visible and easy to adjust.

Building on those preferences, my thematic analysis of the user testing revealed both expected outcomes but also surprising insights into how participants engaged with the brush.

Using a combination of deductive and inductive coding, I analyzed behavioral patterns and direct feedback from five users who interacted with the tool. The themes that emerged provide a clearer picture of what works, what doesn't, and how users adapted to my nontraditional pointing technique.

From a deductive standpoint, several core design goals were validated. The hover delay was able to successfully reduce accidental selections, particularly in areas with densely packed targets, and the visual trail was mentioned for being both satisfying and offering real-time feedback, making users feel in control. The right-click deselection feature, while implemented to improve error recovery, was utilized. Additionally, the combination of being in 'brush mode', after selecting CTRL and the mouse press activation, prevented unintended use.

Quantitatively, the results reinforced these observations. The average correct selection rate across trials was 89%, while the net accuracy, which accounted for incorrect selections, hovered around 75%. Most users made fewer than 3 deselections per trial, and the deselection-to-selection ratio averaged under 15%, indicating that while mistakes occurred, users were often able to correct them with ease. Only a handful of incorrect selections remained at the end of trials, suggesting that the combination of feedback, delay, and undo options helped minimize persistent errors.

On the inductive side, my findings highlighted unexpected behaviors and challenges. One user expected the brush to work by simply hovering, not dragging, indicating a lingering bias toward traditional hover-based interactions. Others used the brush like a highlighter or stylus, valuing the fluidity and "satisfying" trail effect that would fade as they dragged the mouse around. Interestingly, the auto-grouping behavior received mixed reactions: while some users found it helpful for quickly selecting clusters of targets, others found it unpredictable when they

intended to select only one. This suggests that users want more control over selection granularity, perhaps through adjustable sensitivity or grouping toggles.

In conclusion, I believe the key takeaway from my brush design and user study is that users value a balance between control and efficiency. Features like the hover delay, visual trail, and right-click deselection were effective in reducing errors and boosting confidence. However, the study also highlighted areas for improvement—particularly around discoverability, grouping precision, and mode activation clarity. In a future version, I would introduce adjustable grouping sensitivity, a clearer onboarding prompt for brush activation, and stronger real-time feedback on selections and deselections to further improve usability and adaptability across user types.